
Exercise 1

1 Understanding LQR

Consider the following LQR formulation

$$\begin{aligned} \min_u \quad & \sum_{k=0}^{N-1} x_k^T Q x_k + u_k^T R u_k + x_N^T P x_N, \\ \text{s.t.} \quad & x_{k+1} = A x_k + B u_k, \forall k \in [N-1]. \end{aligned}$$

1. Explain how the controller will behave when 1) $Q = 0, P = 0$. 2) $R = 0$.

If $Q = 0, P = 0$, the cost function becomes $\min_u \sum_{k=0}^{N-1} u_k^T R u_k$, which is only related to the inputs. The optimal solution to this problem is $u_k = 0, \forall k \in [N-1]$.

If $R = 0$, the objective becomes $\min_u \sum_{k=0}^{N-1} x_k^T Q x_k + x_N^T P x_N$, which is not related to the control inputs u . The system will settle to $x = 0$ as fast as possible.

2. Explain the difference between finite horizon LQR and infinite horizon LQR ($N = \infty$). Why is the infinite horizon controller time invariant?

For finite horizon LQR, the controller is $u_k = -K_k x_k$, where K_k is obtained by solving the discrete riccati equation. For infinite horizon LQR, $\lim_{k \rightarrow \infty} K_k = K$. The controller becomes $u_k = -K x_k$ with K being a constant matrix. Hence, the infinite horizon controller is time-invariant.

2 Dynamic programming

Consider the following optimal control problem

$$\begin{aligned} V^*(x_0) = \min_u \quad & \gamma x_N^2 + \sum_{k=0}^{N-1} u_k^2, \\ \text{s.t.} \quad & x_{k+1} = x_k + u_k, \end{aligned}$$

where only the cost of the final state is penalized with $\gamma \geq 0$. Your goal is to solve this problem with dynamic programming.

1. Assume that $V_{k+1}(x_{k+1}) = h x_{k+1}^2$ and derive $V_k(x_k)$.

The Bellman recursion is

$$V_k(x_k) = \min_{u_k} u_k^2 + V_{k+1}(x_{k+1}) = \min_{u_k} u_k^2 + h(x_k + u_k)^2.$$

Take the derivative of $V_k(x_k)$ with respect to u_k , we have

$$u_k + h(x_k + u_k) = 0,$$

and

$$u_k = -\frac{h}{1+h} x_k.$$

Plug u_k back in $V_k(x_k)$, we obtain

$$\begin{aligned} V_k(x_k) &= \frac{h^2}{(1+h)^2} x_k^2 + \frac{h}{(1+h)^2} x_k^2 \\ &= \frac{h}{1+h} x_k^2. \end{aligned}$$

2. What is $V_N(x_N)$?

$$V_N(x_N) = \gamma x_N^2$$

3. Compute the optimal value function $V^*(x_0)$ and the control law $\kappa^*(x_0)$ for a finite horizon N .

Starting from $V_N(x_N)$, we know

$$V_{N-1}(x_{N-1}) = \frac{\gamma}{\gamma + 1} x_{N-1}^2,$$

$$V_{N-2}(x_{N-2}) = \frac{\gamma}{2\gamma + 1} x_{N-2}^2,$$

$$V_{N-3}(x_{N-3}) = \frac{\gamma}{3\gamma + 1} x_{N-3}^2.$$

Hence, $V^*(x_0) = \frac{\gamma}{N\gamma + 1} x_0^2$. The optimal control law $\kappa^*(x_0) = -\frac{h_1}{h_1 + 1} x_0 = -\frac{\gamma/((N-1)\gamma + 1)}{\gamma/((N-1)\gamma + 1) + 1} x_0 = -\frac{\gamma}{N\gamma + 1} x_0$.

4. Compute the closed-loop pole of your system. Will your system converge to zero for all finite N and γ ?

The closed-loop system is

$$x_{k+1} = x_k + \kappa(x_k) = \frac{(N-1)\gamma + 1}{N\gamma + 1} x_k.$$

Hence, the closed-loop pole is $\frac{(N-1)\gamma + 1}{N\gamma + 1}$. The system will converge to zero for all finite N and γ since the pole is smaller than one.

5. What is your control law when $N \rightarrow \infty$. Will your system converge to zero for infinite N and finite γ ?

$\lim_{N \rightarrow \infty} \kappa(x_k) = 0$. That means the control inputs will always be zero. The pole of the closed-loop system converges to one. Hence, the system will not move.